Probabilistic dengue and chikungunya forecasting in Brazil: metric-dependent model rankings, regime-shift fragility, and conformal ensembles

Md Kamruzzaman Kamrul[1], Eashraque Jahan Easha[2], and Abdullah Al Helal[3]

[1]Purdue University, West Lafayette, IN, USA
[2]University of Denver, Denver, CO, USA
[3]Oklahoma State University, Stillwater, OK, USA

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Abstract

Dengue is expanding across Brazil; 2024 produced a record epidemic roughly four times any prior year. Accurate, calibrated probabilistic forecasts support preparedness, but which of statistical, machine-learning, deep-learning, and mechanistic approaches forecasts best, especially under an outbreak year's distribution shift, is unsettled. We report a reproducible, leakage-controlled comparison of five probabilistic forecasting model families for the 3rd Infodengue–Mosqlimate Dengue Challenge (IMDC 2026), which requires weekly, state-level, full-interval forecasts for the 26 mandatory Brazilian states. All models are evaluated through a single backtesting harness over four official retrospective seasons, one of which contains the 2024 outbreak, and scored by the weighted interval score (WIS). We report four results with implications beyond this challenge. First, no model is best across seasons: a global recurrent network is most accurate in normal seasons yet worst on the 2024 outlier, while an ensemble never incurs any member's worst-season failure. Second, the best model depends on the evaluation metric: magnitude-weighted mean WIS and scale-free normalized WIS reward accuracy on different subsets of a highly skewed set of states. Third, a conformal recalibration, a post-hoc widening of the ensemble's intervals tuned to fix their overconfidence, improves accuracy and calibration at negligible cost (mean WIS, a magnitude-weighted score, 1234 to 1162; normalized WIS, its scale-free counterpart, 0.595 to 0.561; out-of-sample reduction of 6.2 percent), acting as asymmetric insurance: little cost in an ordinary season, but it prevents the catastrophic under-prediction an extreme one would otherwise cause. Fourth, added model complexity did not pay: observed-climate covariates and hierarchical spatial pooling both failed to improve forecasts, and an internal-holdout hyperparameter gain reversed on the true backtest. The best model is disease-specific: for chikungunya, gradient boosting alone outperforms the ensemble. All code, the backtesting harness, per-state scored predictions, and validated submissions for four official tracks are released, with an automated test suite covering leakage, scoring, and every model.

Author summary

Brazil carries one of the world's largest dengue burdens, and 2024 brought an epidemic without precedent in the national record. Public-health agencies increasingly rely on forecasts that give a full range of plausible outcomes with calibrated uncertainty, not just one predicted number. Which method to trust is an open question, especially in an exceptional season, when accurate forecasts matter most. We compared five model families, from simple seasonal baselines to gradient boosting, deep learning, and a mechanistic epidemic model, using identical retrospective tests across four dengue seasons in all 26 mandatory Brazilian states. No method is best everywhere: the deep-learning model was most accurate in ordinary seasons but failed badly during the 2024 outbreak, while a simple ensemble that averages the models was the most dependable. The apparent winner also changes depending on how performance is scored, a caution for forecasting leaderboards. Finally, a light statistical recalibration that stretched the ensemble's prediction intervals just enough to fix their overconfidence improved both accuracy and reliability. Adding more data or model complexity did not help. At the challenge's own validation-round results presentation, our submission was independently named one of six best-overall-performing models (of 27 submitted) for the mandatory state-level dengue target, and one of three models with outstanding performance on the optional city-level target [34]. Our full pipeline is openly available.

1. Introduction

Dengue is among the fastest-spreading mosquito-borne diseases, with an estimated 390 million infections each year and roughly half of the world's population at risk, a range that continues to expand with climate change and urbanization [25, 27, 28]. Brazil bears one of the largest national burdens [16]. The 2024 season broke the national record by a wide margin, with a peak of 451,578 reported cases in a single epidemiological week and an annual total near 6.67 million, roughly four times the next-largest year on record. Transmission is shaped by climate, and the 2023–24 El Niño is one of several drivers implicated in the surge. Against this background, arboviral forecasting has moved from point prediction toward full probabilistic forecasts, motivated by the recognition that preparedness decisions depend on the entire predictive distribution rather than a single expected value [17, 21, 30].

The Infodengue–Mosqlimate Dengue Challenge (IMDC) operationalizes this shift. It invites teams to submit weekly, state-level probabilistic forecasts with prescribed quantiles, scored by the weighted interval score, following the precedent of the inaugural 2024 edition [1]. The challenge exposes a question that the wider forecasting literature has not resolved for this setting: given statistical, machine-learning (ML), deep-learning (DL), and mechanistic approaches, which forecast best, and how robustly, when a season can depart by an order of magnitude from anything in the training record?

Prior work on dengue forecasting is extensive but leaves that question open in three specific ways. First, most published comparisons rank models on point accuracy. A systematic review and network meta-analysis of 59 dengue-forecasting studies from 2014 to 2024 ranked methods on root-mean-square error and found simple statistical models competitive with machine learning at lower data and implementation cost [6]. Point accuracy does not measure whether a prediction interval is calibrated, which is what a preparedness decision needs. Second, the probabilistic comparisons that do exist mostly target short horizons. The largest, an evaluation across more than 180 locations worldwide, forecasts one to three months ahead and finds ensembles consistently among the top performers [3]; a Brazilian statistical model forecasts probabilistic epidemic bands, the activity range a coming season is expected to fall within, rather than a weekly trajectory [4]. A full-season target, here 16 to 67 weeks ahead because a 15-week reporting gap separates the training cutoff from the first target week, is a different problem: no recent case data is available near the target, so the models that win at one month need not win here. Third, the inaugural sprint reported that no single model excelled across locations and years, but did not isolate why rankings move, nor test whether the choice of aggregate metric itself changes the winner [1].

We address this with a controlled comparison built for reproducibility and for honest evaluation under distribution shift. Every model is fit and scored through one backtesting harness that enforces leakage safety and applies the same scoring code to all forecasts, over the four official retrospective seasons (2022–23 through 2025–26) for the 26 mandatory states. The design trades breadth for depth relative to multi-country evaluations [3]: one country, two diseases, a full-season horizon, and every model sharing one leakage-audited pipeline whose tests, provenance manifest, and per-state scored predictions are public. That depth is what makes the negative results and the metric-disagreement finding below checkable rather than anecdotal. We make the following contributions.

1. We show that model rankings are season-dependent and, in the outbreak year, invert: the model that is best on average can be the least reliable exactly when reliability matters most (Section 2.2).

2. We show that the identity of the best model depends on whether skill is weighted by case magnitude or by forecasting unit, a consequence of real cross-state heterogeneity rather than an artifact, and we argue for reporting both (Section 2.5).

3. We introduce a conformal recalibration of the ensemble intervals that improves accuracy and calibration at negligible cost and functions as asymmetric insurance against catastrophic under-prediction (Section 2.3).

4. We document informative negative results: added covariates, hierarchical pooling, hyperparameter tuning, and an alternative ensemble-combination rule each failed to improve forecasts where it mattered most, which clarifies where the predictable signal in this problem lies (Section 2.8).

5. We show that the best model is disease-specific, using chikungunya as a contrasting target (Section 2.9), and we release the complete, tested pipeline.

2. Results

2.1. Study design and data

We forecast weekly dengue case counts per state across the 2022–23 through 2025–26 seasons for the 26 mandatory Brazilian states (Espírito Santo is excluded per the challenge specification). These four seasons are referred to throughout as folds 1 through 4 (fold 1: 2022–23; fold 2: 2023–24, containing the 2024 outbreak; fold 3: 2024–25; fold 4: 2025–26, partially resolved). Each fold

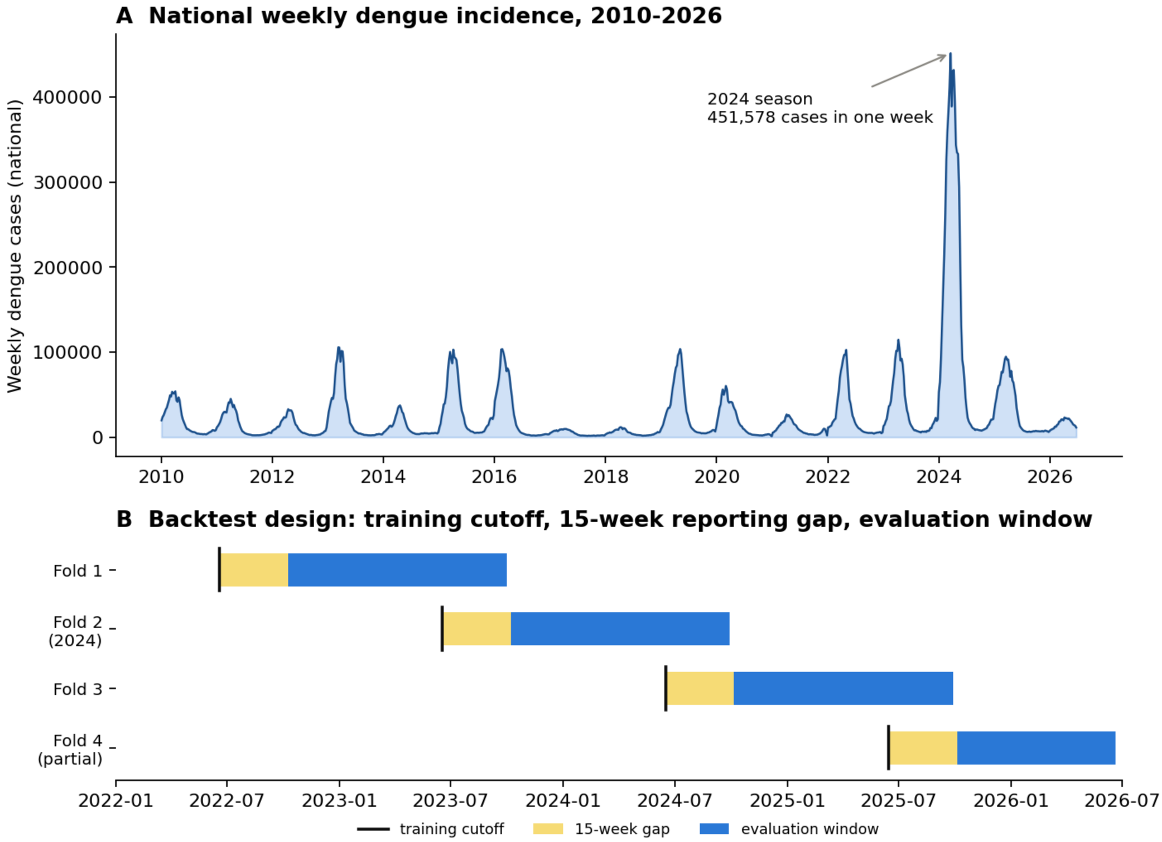

Figure 1: Data and study design. (A) Weekly reported dengue cases aggregated to the national level, 2010–2026; the 2024 season stands roughly four times above any prior year. (B) The four official retrospective seasons form a rolling backtest. Each has a training cutoff (all data up to the cutoff is used for fitting), a 15-week reporting gap, and an evaluation window on which forecasts are scored. Fold 2 covers the 2024 outbreak and fold 4 is partially resolved.

is defined by an official training cutoff, the last date a model fitted for that fold is allowed to see, and a target window, the season it must forecast; a confirmed 15-week gap separates each training cutoff from the start of its evaluation window, reflecting the reporting delay a real forecaster faces (Fig 1). Forecasts are submitted as nine quantiles per state and week, from which the four central prediction intervals (50, 80, 90, and 95 percent) are formed. All covariate access, feature construction, and model fitting are filtered to data at or before each training cutoff, and this discipline is enforced by automated tests rather than by convention (Section 4). The four seasons differ by an order of magnitude in difficulty. The 2023–24 season (fold 2), which contains the 2024 outbreak, is the hardest for every model, and the 2025–26 season (fold 4) is only partially resolved in the available data and is reported separately.

2.2. No single model dominates across seasons

Table 1 reports mean WIS by model and season, and Fig 2 shows the same on a log scale. The ranking of models reorders across seasons. The deep network (a global GRU with a negative-binomial head, an output layer suited to over-dispersed count data such as case counts, rather than the Poisson head more common in count models) is the single best model in the two ordinary seasons, folds 1 and 3 (WIS 348 and 324), but is the worst of all models on the 2024 outlier, fold 2 (WIS 4131), worse even than a naive baseline, because the surge lies outside its training distribution and

Table 1: Dengue mean WIS by season (fold). Lower is better. Fold 2 contains the 2024 outbreak and is roughly ten times harder than any other season. Fold 4 (2025–26) is partially resolved and reported separately. Best value in each column is in bold.

Model	Fold 1 2022–23	Fold 2 2023–24	Fold 3 2024–25	Fold 4 2025–26
Ensemble (conformal)	337	3297	403	408
Ensemble (Vincentization)	338	3574	398	399
Ensemble (inverse-WIS)	**313**	3625	**387**	401
GRU (deep learning)	348	4131	**324**	542
XGBoost	376	**3238**	484	468
Mechanistic	468	3371	551	314
Climatological	386	3588	516	309
Seasonal-naive	500	3848	581	293
Naive	726	3833	1056	**266**

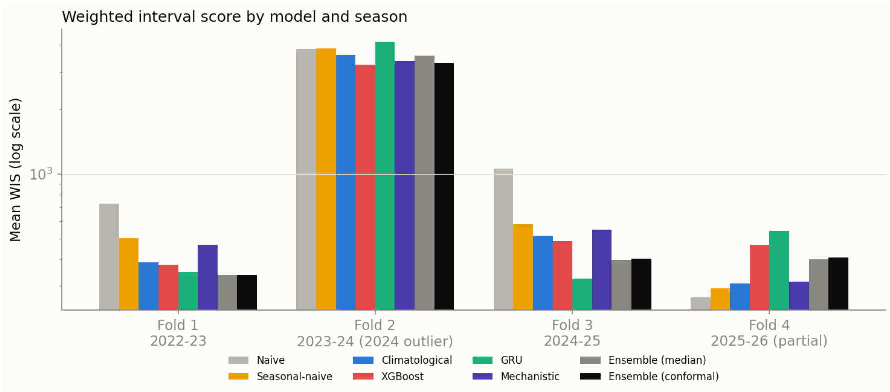

Figure 2: Mean WIS by model and season (log scale). Model rankings reorder across seasons. The deep network is best in ordinary seasons and worst on the 2024 outlier; the ensemble avoids every model's worst-season failure. The conformal-recalibrated ensemble (Section 2.3) is shown alongside its members.

its intervals, derived from its fitted distribution rather than built empirically, are too narrow to cover it. The gradient-boosted model (XGBoost, chosen over a second gradient-boosting implementation, LightGBM, that we also tested and that scored significantly worse overall, Section 4) is the strongest single model on the outlier season (WIS 3238), but unlike LightGBM, which showed a sharper trade-off (strongest on the outlier at WIS 3332 yet the weakest of the serious models on the ordinary season fold 3, WIS 665), XGBoost remains reasonably competitive rather than collapsing in the ordinary seasons (WIS 376 and 484 in folds 1 and 3). The semi-mechanistic trajectory model is near best on the outlier (WIS 3371) because its bootstrap of historical seasons yields wide intervals precisely when they are needed. This season-dependence is the central obstacle the challenge poses: a model selected on average performance carries a hidden risk of catastrophic failure in the season that matters most.

Season-to-season stability is a distinct axis from mean skill, and the two disagree for the models identified above. For each fold we compute each model's relative WIS against the seasonal-naive baseline (a fully-crossed pairwise comparison over state and horizon, Cramer et al. 2022), take its log so that a model twice as good and a model twice as bad are symmetric, and summarize the four folds by their mean (overall skill against the baseline, negative is better) and standard deviation (season-to-season consistency, lower is more stable). Fig 3 plots the two against each other. A log relative WIS of -0.42, for instance, means a model whose WIS is on average $e^{0.42} \approx 1.5$ times

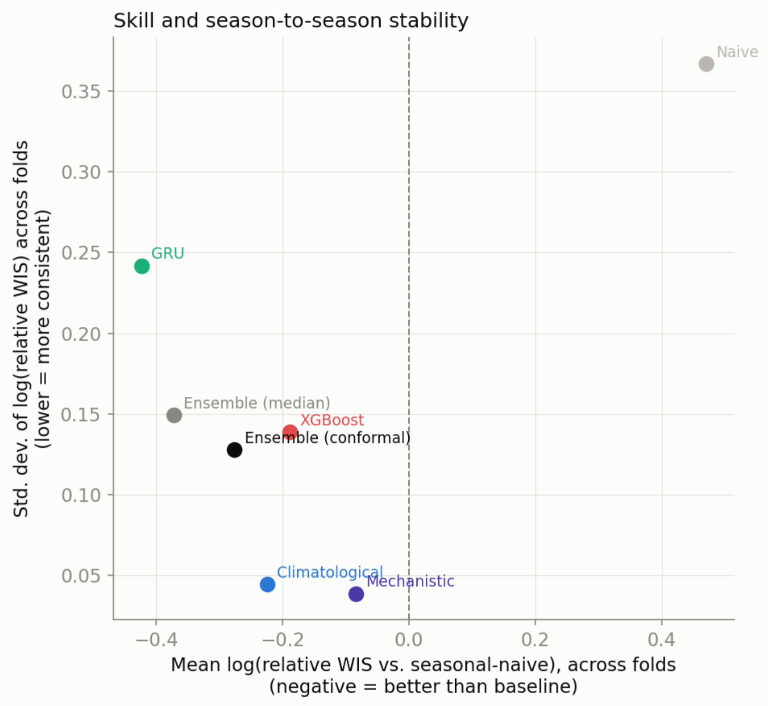

Figure 3: Skill and season-to-season stability are distinct axes. Each point is one model's mean log relative WIS against the seasonal-naive baseline across the four folds (x, negative is better) against the standard deviation of that same quantity (y, lower is more consistent). The deep network has the best mean skill but is among the least stable; the mechanistic model is the most stable but has only modest mean skill; the ensembles occupy an intermediate position that no single member reaches.

smaller than the baseline's, i.e. about 1.5 times more accurate; a positive value of the same size would mean 1.5 times worse. The deep network has the best mean skill of any model on this scale (-0.42) but also the highest variability of any model except naive (0.24), the quantitative signature of its ordinary-season strength and outlier-season failure. The mechanistic model sits at the opposite corner: modest mean skill (-0.09) but the lowest variability of any model (0.04), because its bootstrap of historical seasons widens intervals in proportion to historical variability rather than assuming a fixed noise scale. Both ensembles trade part of the deep network's mean skill for markedly better consistency (std 0.15 and 0.13 for the median and conformal ensembles), which is the quantitative counterpart of the qualitative claim above: an ensemble is not chosen for being the single best model in any one season, but for never being the worst.

2.3. An ensemble is the most robust model, and conformal recalibration makes it the most accurate

Because different models win different seasons, a combination that is never worst is attractive. An unweighted per-quantile median ensemble, Vincentization, meaning that at every quantile level (the median, the 2.5th percentile, and so on) we take the median of the three models' values at that level, of the climatological, gradient-boosted (XGBoost), and deep-learning models avoids the catastrophic single-season failure of its weakest member (WIS 3574 in the outlier season, far below the deep network's 4131) but, once XGBoost replaced a weaker gradient-boosted alternative as the strongest single model (Section 4), no longer beats that single model outright on raw WIS alone (1234 versus 1190 for XGBoost; Table 2). Its interval calibration is close to nominal but slightly

Table 2: Overall dengue backtest. Averaged over four seasons and 26 states. Mean WIS is magnitude-weighted; normalized WIS (sum of WIS divided by sum of cases) is the challenge's headline metric and is reported for all folds and excluding the 2024 outlier. Coverage nearer the nominal 50/80/90/95 is better. Best value in each column is in bold.

Model	WIS	nWIS	nWIS	Coverage (%)			
		all	ex-2024	50	80	90	95
Ensemble (conformal)	**1162**	**0.561**	0.375	46	74	87	94
XGBoost	1190	0.575	0.434	48	79	88	92
Ensemble (Vincentization)	1234	0.595	0.371	43	72	82	86
Ensemble (inverse-WIS)	1238	0.598	**0.358**	44	74	84	89
Mechanistic	1238	0.598	0.450	55	80	86	90
Climatological	1264	0.610	0.407	48	74	83	86
LightGBM	1301	0.628	0.549	48	80	89	93
Seasonal-naive	1379	0.666	0.468	43	78	87	92
GRU (deep learning)	1394	0.673	0.385	27	50	60	67
Naive	1557	0.752	0.713	41	73	81	87

narrow.

We improve on it with a conformal recalibration of the ensemble intervals, a distinct step from the conformalized quantile regression (CQR) used inside the gradient-boosted model (Section 4.6 gives the full procedure): both stretch prediction intervals to fix them being too narrow, but CQR calibrates one model's own intervals during fitting, whereas this step recalibrates the already-combined ensemble's intervals afterward. Calibration factors that make each central interval reach its nominal coverage are estimated on a single tuning season (fold 1) and applied to all seasons. The estimated factors are gentle and concentrate on the tails: 1.08, 1.04, 1.19, and 1.86 for the 50, 80, 90, and 95 percent intervals. Concretely, a raw 95 percent interval of, say, [40, 160] around a median of 100 becomes approximately [−12, 212], clipped to [0, 212] for non-negativity, after the 1.86 factor: the median is never touched, only the interval stretches, and it stretches only where it needs to (the 80 percent interval above needed only a negligible correction). This step is what turns the ensemble into the single best model overall, not merely a hedge against any one member's worst case: the recalibrated ensemble is the most accurate model overall (mean WIS 1162 versus 1234 for the plain ensemble and versus 1190 for the best single model, XGBoost; normalized WIS 0.561 versus 0.595) and better calibrated at the tails (empirical coverage 46/74/87/94 versus 43/72/82/86), as shown in Fig 4. Out of sample, on the three seasons not used for calibration, the reduction is 6.2 percent. The gain is concentrated in the outbreak season (fold 2 WIS 3574 to 3297) at a small cost in ordinary seasons, so the recalibration behaves as asymmetric insurance: it widens the outer tails just enough to blunt catastrophic under-prediction in an extreme year while barely affecting normal seasons. Recalibrating the ensemble output outperforms recalibrating any single member, because the per-quantile median across members otherwise dilutes a single member's widening.

2.4. The key differences are statistically significant

The absolute skill of each model carries wide uncertainty. A block bootstrap, repeatedly resampling whole state-season units (with replacement) and recomputing normalized WIS on each resample to see how much the result varies, over the 104 state-season units gives a 95 percent confidence

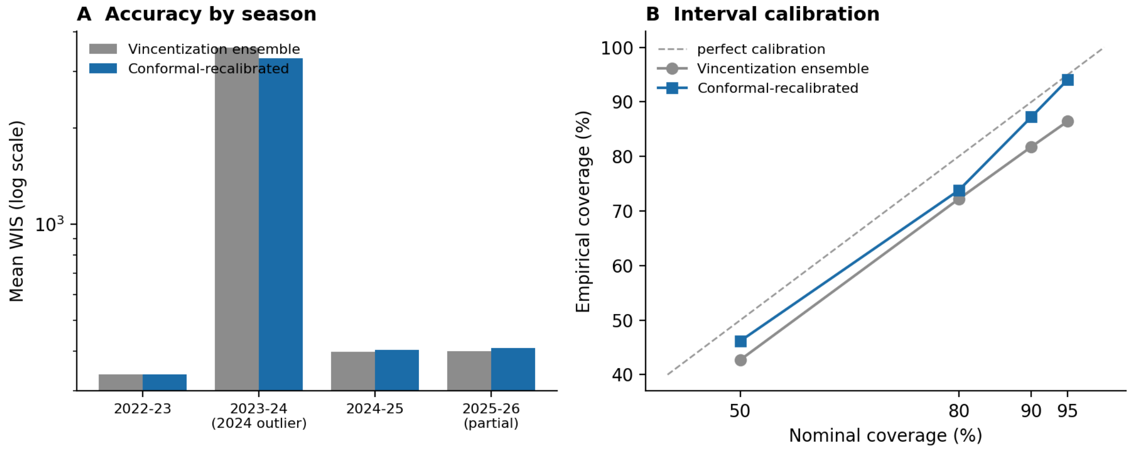

Figure 4: Effect of conformal recalibration on the dengue ensemble. (A) Mean WIS by season, Vincentization ensemble versus the conformal-recalibrated ensemble (log scale). The gain concentrates in the 2024 outlier season at a small cost in ordinary seasons. (B) Interval calibration: empirical versus nominal coverage. Recalibration moves the outer intervals (90, 95) toward the diagonal.

interval (CI) on the conformal ensemble's normalized WIS of $[0.423, 0.639]$, and the intervals of the leading models overlap (Fig 5A), a direct consequence of the outbreak season inflating the variance of any magnitude-weighted score. The comparisons that matter are nonetheless decisive, because the models are strongly correlated across units, meaning they tend to do well or badly on the same state-seasons, so a paired analysis that looks at each model's difference from another on the very same units cancels that shared, season-driven variance instead of drowning in it (Fig 5B). The conformal recalibration improves on the median ensemble by a mean 71.3 WIS (95 percent CI $[56.9, 87.8]$) and on the best single model, XGBoost, by 27.9 ($[8.8, 47.3]$); on ordinary seasons the deep network beats XGBoost by 48.9 ($[14.9, 87.4]$); XGBoost itself beats the alternative gradient-boosted implementation we also tested, LightGBM, by 111.1 ($[81.6, 143.0]$); and the conformal ensemble beats the naive baseline by 395.1 ($[338.8, 457.3]$). All five intervals exclude zero. The lesson is methodological as much as substantive: in a record with one extreme season, marginal skill intervals are wide, but paired comparisons remain informative and are the sound basis for model selection.

2.5. The winner depends on the evaluation metric

The overall mean WIS in Table 2 is dominated by the largest states and by the outbreak season, both of which contribute large absolute scores. The challenge's headline metric is instead a normalized WIS, the sum of WIS divided by the sum of observed cases, which is scale-free across regions. Under this metric the ordering changes in a revealing way. The conformal ensemble is best over all folds (normalized WIS 0.561), consistent with the mean-WIS ranking. When the 2024 outlier is excluded to isolate ordinary-season skill, the deep network rises from worst of six leading models by mean WIS to third-best by normalized WIS (0.385), and the plain median ensemble overtakes the conformal ensemble for the single best spot on this metric (0.371 versus 0.375), while the gradient-boosted model (XGBoost) falls from second-best to fifth of six (0.434) and the mechanistic model falls from fourth to last (0.450) (Fig 6). The reversal is real but milder for the gradient-boosted model than it was for the LightGBM configuration we tested first, which fell all the way to the single weakest of the serious models under this metric (0.549, still its value in

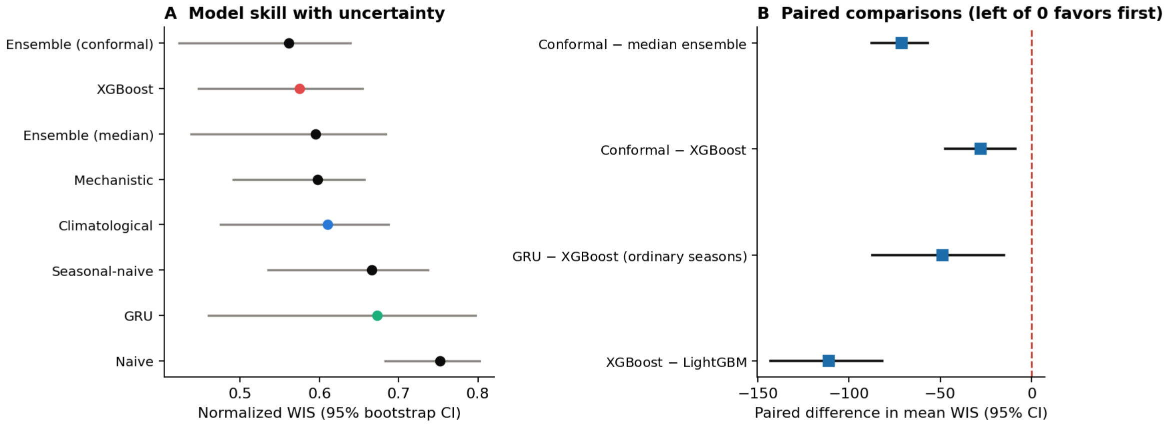

Figure 5: Statistical significance of the model comparisons. (A) Normalized WIS with 95 percent block-bootstrap intervals over state-season units; the leading models' intervals overlap because the 2024 outbreak inflates the variance of absolute skill. (B) Paired differences in mean WIS with 95 percent intervals; all four comparisons lie left of zero, so the conformal ensemble significantly beats the median ensemble and XGBoost, the deep network significantly beats XGBoost on ordinary seasons, and XGBoost significantly beats the alternative gradient-boosted implementation we also tested, LightGBM.

Table 2 since LightGBM itself was not adopted): XGBoost's more balanced profile across seasons (Section 2.2) narrows, but does not eliminate, the metric-dependent reordering that motivates reporting both metrics. The disagreement is not a paradox. Consider two toy states, one with 10,000 weekly cases and one with 100: a model that misses the large state's count by 1,000 and the small state's by 50 has a much larger magnitude-weighted error, dominated by the 1,000, than a metric that instead scores each state's error relative to its own size (a 10 percent miss on the large state versus a 50 percent miss on the small one, so a scale-free metric would actually rate the small state's forecast as worse). A metric that rewards accuracy on a few very large states favors different models than one that rewards accuracy on the many small states, and the 2024 season inflates the mean for whichever model handles it least badly. We therefore report both magnitude- and unit-weighted skill, and we caution that an epidemic-forecasting leaderboard reporting a single aggregate can mislead. The cross-state skew has a clear geographic structure (Fig 7): forecast difficulty, measured by normalized WIS, is greatest in the dense, high-incidence southeastern states and lowest in the low-incidence interior, so magnitude-weighted skill is determined by a few states.

2.6. Calibration and its correction

Raw gradient-boosted and deep-learning quantile models are over-confident (Fig 8): their nominal coverage, the fraction of outcomes an interval is supposed to contain, e.g. 90 percent for the 90 percent interval, is higher than their empirical coverage, the fraction it actually does contain. The GRU's nominal 50 percent interval only actually contains the truth 27 percent of the time, and its nominal 95 percent interval only 67 percent of the time, and it is under-covered in every season rather than only in the outbreak year, which identifies the problem as systematic over-confidence rather than a regime-specific failure. Conformalized quantile regression restores the gradient-boosted model (XGBoost) to near-nominal calibration (48/79/88/92) at negligible WIS cost. The same correction applied to the deep network trades away sharpness, how narrow and therefore informative its intervals are, on the seasons where it is already accurate, so calibration and sharpness trade off at the level of a single model. The per-quantile median of the ensem-

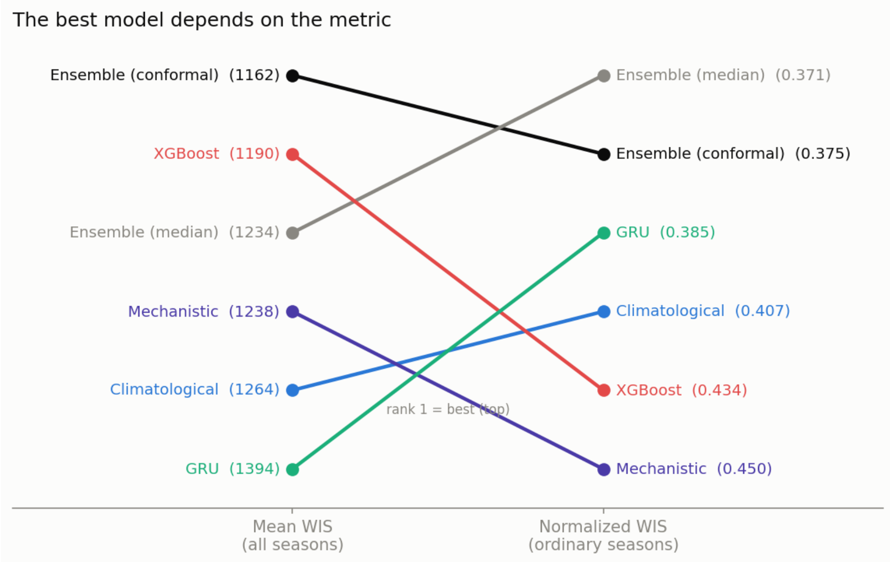

Figure 6: The best model depends on the metric. Model rank by magnitude-weighted mean WIS over all seasons (left) and by normalized WIS over the ordinary seasons with the 2024 outlier excluded (right); rank 1 is best, among these six models. The rankings reorder: the deep network rises from last by mean WIS to third on ordinary seasons, the gradient-boosted model (XGBoost) falls from second to fifth, the mechanistic model falls from fourth to last, and the plain median ensemble overtakes the conformal ensemble for the top spot. Values in parentheses are mean WIS (left) and normalized WIS (right).

ble provides a structural calibration improvement at no cost, and the conformal recalibration of Section 2.3 completes it.

2.7. Skill varies with horizon, and the outbreak failure is under-prediction

Because the 15-week reporting gap places every target 16 to 67 weeks ahead, we stratified skill by lead time on the ordinary seasons (Fig 9A). The deep network's advantage is horizon-dependent: it is the best model at short lead times (normalized WIS 0.255 at 16 to 27 weeks) but degrades at the longest horizons (0.388 at 52 to 67 weeks), where both the conformal ensemble (0.338) and the gradient-boosted model (0.349) overtake it. The conformal ensemble is the most stable model across horizons and is never worst, which mirrors its stability across seasons. WIS decomposes exactly into three non-negative components (Section 4.4): dispersion (a penalty for interval width, present even when the forecast is centered correctly), under-prediction, and over-prediction; decomposing it this way explains the outbreak-season failure directly (Fig 9B): in the 2024 season the ensemble's score is dominated by under-prediction (a mean of 2675 WIS units, against 621 for dispersion and 1 for over-prediction), whereas every other season is dominated by dispersion, i.e. is driven by ordinary interval width rather than the forecast being systematically off in either direction. The record epidemic was not mis-shaped but simply larger than any model would predict from pre-season information, which is why widening the upper tail, as the conformal recalibration does, is the effective remedy.

2.8. Added complexity did not improve forecasts

Four interventions that plausibly should have helped did not (Table 3). First, adding observed-climate covariates to the gradient-boosted model, following the distributed-lag non-linear (DLNM)

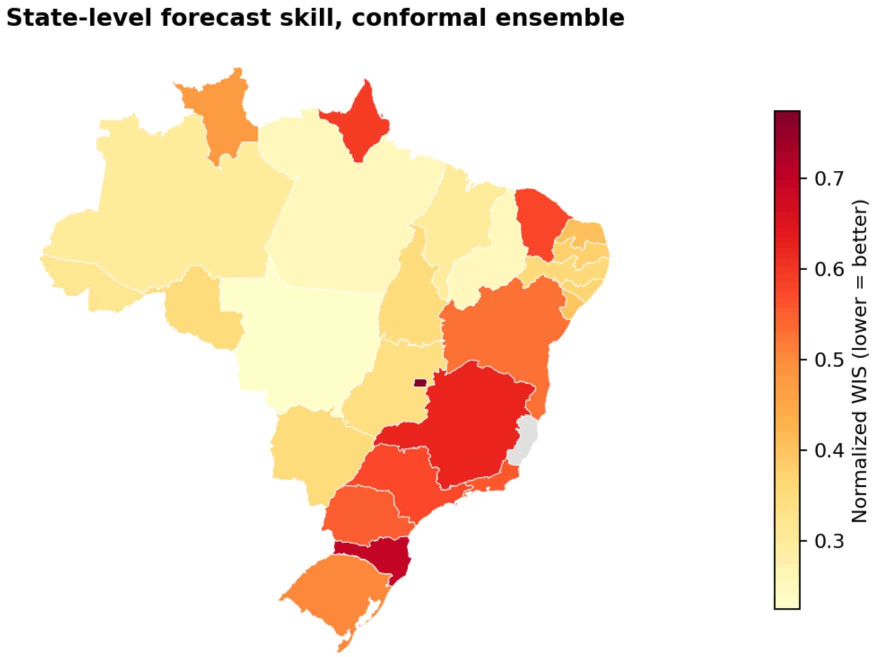

Figure 7: Geographic structure of forecast skill. Normalized WIS of the conformal ensemble by state (lower is better). Forecasting is hardest in the dense, high-incidence southeastern states and in the Federal District, and easiest in the low-incidence interior, which is why magnitude-weighted skill is dominated by a few states. Espírito Santo (grey) is excluded from the challenge.

structure of the strongest model in the 2024 sprint, which lets a covariate's effect on cases be both delayed (this week's temperature can affect cases several weeks later, not just this week) and non-linear [1, 15], worsened backtest WIS by 3.3 percent (1299 to 1342 on the LightGBM configuration on which this test was originally run[1]). A feature-importance analysis showed why: the nine added covariates (minimum temperature, a derived standardized-precipitation drought index at one, three, and six month accumulations, multi-month temperature lags, thermal range, and rainy days) together carried only 5.3 percent of importance and ranked below the existing features. The climate signal the model actually uses is the seasonal climate forecast and the El Niño–Southern Oscillation indices, both already present and both forward-looking, rather than

[1]This test and the hyperparameter-tuning test below were originally performed on the LightGBM-based pipeline before it was superseded by XGBoost (Section 4) and before a later data refresh. Re-verified on the current XGBoost-based pipeline and current data (2026-09-09): the climate-covariate finding replicates in the same direction (WIS 1166 → 1190 without vs. with the covariates, a 2.0 percent degradation from adding them, versus the original 3.3 percent). The hyperparameter-tuning finding does *not* replicate: on XGBoost, the config an internal fold-1 holdout preferred (shallower trees, `max_depth` 3 vs. the deployed default 5, with stronger leaf-size regularization) also won the true 4-fold backtest on every single fold (WIS 1190 → 1152, a genuine 3.2 percent improvement, not an overfit to the holdout) – the opposite of the LightGBM-era finding described in the main text below. We report both results rather than silently replacing the original figures, since the discrepancy is itself informative: whether an internal-holdout hyperparameter search transfers to the true backtest evidently depends on the specific model family and hyperparameter space searched, not a fixed property of this forecasting task. This XGBoost re-verification was checked at the ensemble level under this project's own fold-1 tuning discipline (the same check used to validate the original LightGBM-to-XGBoost swap) and did *not* pass: combined with climatological_quantile and gru_negbin and conformal-recalibrated, both the shallower-hyperparameter config alone and combined with dropping the climate covariates lose on fold 1 (WIS 347 and 346 respectively, vs. the deployed ensemble's 337) despite winning on every full-aggregate metric – the same overfitting-to-the-reporting-folds pattern this project has already caught in other candidate changes (Section 2.8). Dropping the climate covariates alone does pass fold 1 at the ensemble level (335 vs. 337) but is a mixed result elsewhere (worse raw WIS and normalized WIS all-folds, better normalized WIS excluding 2024) rather than a clean sweep. None of the three candidates were adopted; the deployed ensemble is unchanged.

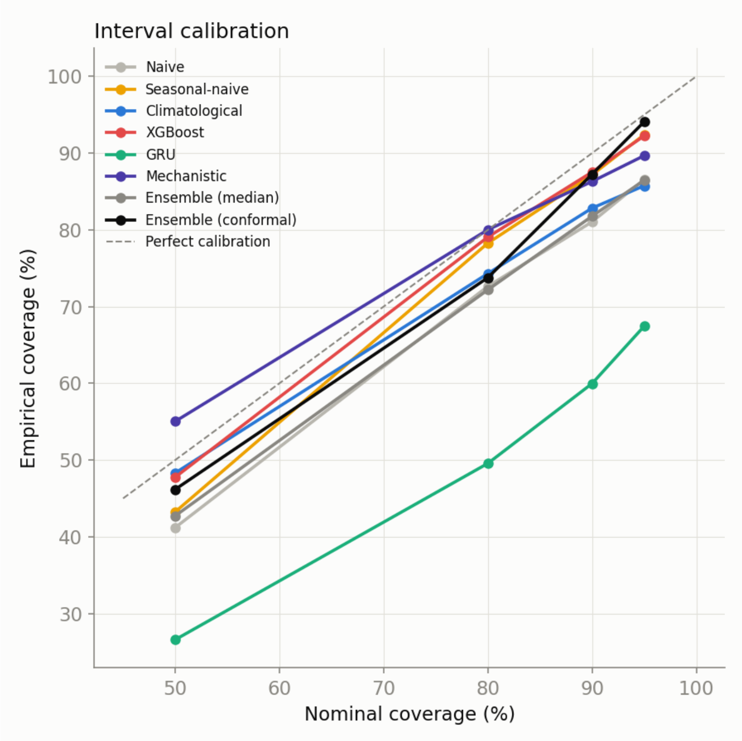

Figure 8: Interval calibration by model. Empirical versus nominal coverage; points on the diagonal are perfectly calibrated. The deep network is markedly overconfident; conformalized quantile regression corrects the gradient-boosted model, and the conformal-recalibrated ensemble tracks the diagonal at the outer intervals.

observed local weather, which is stale at the long horizons that dominate a full-season forecast. Second, a hierarchical, empirical-Bayes pooling of each state's seasonal climatology toward its macroregion, that is, borrowing statistical strength by pulling each state's estimate partway toward its region's average rather than trusting each state's own history entirely, worsened the ensemble from 1162 to 1185, because Brazilian states are too heterogeneous within a macroregion: shrinking a small state toward a region dominated by a larger, differently behaving state introduces bias. Third, a hyperparameter search that improved an internal time-block holdout, a held-out slice of recent training weeks used to tune hyperparameters before touching the real backtest, by 5 percent worsened the true backtest on the LightGBM configuration this was originally tested on, because an internal holdout does not necessarily represent forecasting a full future season from mid-year; conservative, a priori defaults generalized better in that specific test. A re-verification on the current XGBoost-based pipeline found the opposite as a standalone model (see footnote above): the holdout's preferred configuration also won the true backtest there. Checked further at the ensemble level, though, it lost on the same fold-1 tuning check that validated every other change reported here, so it was not adopted – a reminder that a standalone-model result and an ensemble-level result can disagree even when both are individually rigorous. Fourth, we tested the ensemble combination rule itself: the inaugural 2024 sprint's organizers combine models by log-linear (logarithmic) pooling of parametric predictive distributions [1, 2], a genuinely different rule from our per-quantile-median Vincentization – a precision-weighted combination of distributions rather than a per-quantile order statistic, which guarantees a unimodal result. Substituted for Vincentization (equal weights, then conformal-recalibrated identically to the deployed ensemble),

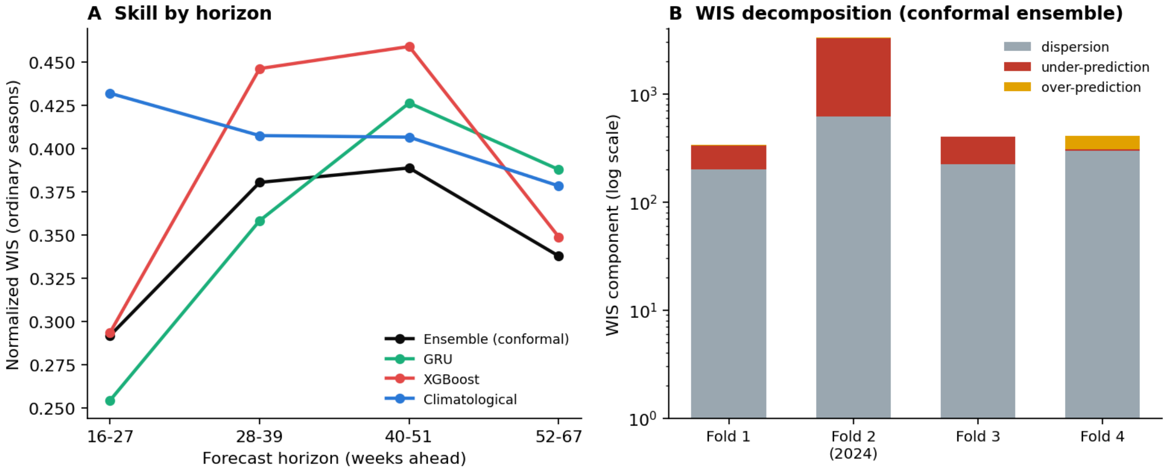

Figure 9: Robustness across horizons and anatomy of the outbreak-season failure. (A) Normalized WIS by forecast horizon on the ordinary seasons; the deep network leads at short lead times and fades at long ones, while the conformal ensemble is stable. (B) WIS decomposition of the conformal ensemble by season (log scale); the 2024 season is dominated by under-prediction, not by dispersion or over-prediction.

it wins three of four folds outright, including fold 1, the only fold this project's discipline permits for tuning decisions (WIS 309 versus 337, an 8 percent improvement) and the ordinary-season metric (normalized WIS excluding 2024, 0.338 versus 0.375). By the letter of that discipline, this candidate passes and every previous one in this table failed. But it loses specifically on the 2024 outbreak fold (WIS 3630 versus 3297, a 10 percent regression, with 90 and 95 percent coverage both falling further from nominal, 64/79 versus 70/83): precision-weighted pooling is sharper in ordinary seasons but narrows the combined tails exactly where the widest tails matter most. We did not adopt it. The fold-1 rule exists to prevent tuning the ensemble to the specific reporting folds it is judged on, not to license a change that measurably weakens performance on the one season the entire ensemble-plus-conformal design is built to protect against; a rule applied mechanically here would have recommended trading away the outbreak-season robustness that motivates this paper's central design choice. Together these results locate much of the predictable signal in this problem, caution against complexity that is not validated on the true forecasting task, and show that even a validation discipline's own conclusions can be implementation-specific – and occasionally require judgment beyond the rule itself – rather than a fixed property of the problem.

2.9. Chikungunya is a more predictable, contrasting target

Brazil reported roughly 422,000 chikungunya cases in 2024, a cumulative incidence near 199 per 100,000, and the virus has recurred across the country since its 2014 introduction with a spatial pattern shaped by heterogeneous population immunity [29]. We forecast chikungunya on the same 26 states, harness, and seasons as dengue. Two differences stand out. First, the gradient-boosted model alone is the best forecaster (WIS 79.1), ahead of the ensemble (89.5) and every other model (Table 4); it is best in every season (Table 5), so combining it with weaker models only dilutes it. Second, chikungunya is markedly more predictable than dengue relative to its scale: the 2024 season is only modestly harder than the others (mean WIS 120 versus a range of 33 to 81 across seasons), in contrast to the order-of-magnitude jump dengue shows in 2024. This matches reports that chikungunya dynamics in Brazil are more regular than dengue's [29], plausibly because immunity

Table 3: Interventions that did not help, plus two that passed part of our validation discipline but still were not adopted (negative results throughout). Each was evaluated on the same backtest as the main comparison. †Originally tested on the LightGBM configuration before it was superseded by XGBoost and before a later data refresh; re-verified on the current XGBoost-based pipeline and current data (see footnote in text). The hyperparameter row's standalone-model finding reversed on re-verification, but checking it at the ensemble level (the same fold-1 discipline that validated every adopted change) found it should not be adopted either. ‡Passes the fold-1 tuning check outright but regresses specifically on the outbreak fold (see text); not adopted on substantive grounds despite passing the formal rule – none of these five rows changed the deployed model.

Intervention	**Effect**	**Reason**
Observed-climate covariates† (XGBoost)	WIS 1166 → 1190 (current); 1299 → 1342 (orig.)	5.3% importance; signal is forecast + ENSO, not observed weather
Hierarchical macroregion pooling (added to ensemble)	WIS 1162 → 1185	small-state bias; too correlated with climatological member
Hyperparameter tuning† (LightGBM, original)	internal holdout +5%, backtest worse	internal holdout misrepresented full-season forecasting
Hyperparameter tuning† (XGBoost)	standalone −3.2%; ensemble fold 1 worse	generalized alone, but not at ensemble level (see text)
Log-linear pooling‡ (vs. Vincentization)	fold 1 −8%; outbreak fold (fold 2) WIS +10%	sharper pooling narrows tails exactly where width matters most

Table 4: Chikungunya backtest. State-level, same harness and seasons as dengue. Gradient boosting alone is best.

Model	**WIS**	**Cov. 50 (%)**	**Cov. 90 (%)**
LightGBM	**79.1**	47	88
Ensemble (Vincentization)	89.5	55	87
Climatological	92.4	56	85
Seasonal-naive	95.1	48	88
Mechanistic	99.5	53	86
Naive	119.1	41	80

accumulates against a single antigenic type rather than interacting across four co-circulating dengue serotypes. The practical consequence is that the best model is disease-specific: a blended ensemble is right for dengue, a single well-calibrated learner for chikungunya, even within one surveillance system and pipeline.

2.10. City-level forecasting is substantially harder

The challenge also solicits municipal forecasts for 15 dengue and 10 chikungunya cities, an optional target we submitted with the climatological quantile model. City-level forecasting is much harder than state-level: the best model's normalized WIS is 0.60 for dengue cities and 0.87 for chikungunya cities (Table 6), against 0.56 and 0.62 at the state level. Interval coverage is also poorer (50 percent coverage of 39 percent for dengue cities), because many mid-sized municipalities have episodic, over-dispersed series with long runs of zero counts that a per-week empirical quantile under-covers. A degenerate all-zero median is a concrete failure mode of these sparse series, which we addressed by switching the point forecast to a clamped predictive mean, the distribution's mean rather than its median, floored at zero so it can never predict a negative case count; a principled zero-inflated treatment is a clear direction for the municipal tracks. These results quantify the resolution penalty of forecasting at the level where interventions are delivered.

Table 5: Chikungunya mean WIS by season. LightGBM is best in every season, unlike dengue, and the 2024 season (fold 2) is only modestly harder than the others.

Model	Fold 1	Fold 2	Fold 3	Fold 4
LightGBM	**81.2**	**120.2**	**69.2**	**33.4**
Ensemble (Vincentization)	88.7	141.1	73.8	41.5
Climatological	94.3	143.3	75.3	43.7
Seasonal-naive	92.8	152.3	77.1	44.6
Mechanistic	92.1	166.6	77.0	48.6
Naive	181.3	145.7	75.3	57.7

Table 6: City-level backtest (optional tracks; climatological quantile model submitted). WIS and normalized WIS averaged over the target cities and four seasons. For chikungunya cities, climatological and seasonal-naive are statistically indistinguishable (WIS 30.17 versus 30.17); no value is bolded there for that reason.

Disease	Model	WIS	nWIS	Cov. 50 (%)
Dengue (15 cities)	Climatological	**123.3**	**0.60**	39
	Seasonal-naive	131.8	0.64	30
	Naive	133.5	0.65	30
Chikungunya (10 cities)	Climatological	30.2	0.87	57
	Seasonal-naive	30.2	0.87	50
	Naive	34.4	1.00	33

2.11. Onset, peak, and total burden are decision-relevant summaries that WIS does not directly show

WIS scores every week of the season equally, but health-system planning asks three narrower questions: when will the season start, how large will the peak be, and how many cases will there be in total. We compute, per state-season unit, the error in the first week each series reaches 10 percent of its own peak (onset), the timing and log-magnitude error at the true peak week (as in Section 2.2), and the log-ratio of predicted to observed total season cases (Table 7). No model wins on every one of these summaries, which is the same qualitative pattern as the main WIS results but on a different, more directly interpretable set of outputs. The gradient-boosted model (XGBoost) has the lowest onset-timing error (3.7 weeks mean absolute error, against 3.8 for the conformal ensemble), the climatological model has the lowest peak-timing error (4.5 weeks, against 4.9 for the ensemble), and the gradient-boosted model again gives the least biased total-season case count (a log-ratio of -0.01, i.e. predicting within 1 percent of the true total at the median unit, against 20 percent under-prediction for the ensemble and 56 percent for the mechanistic model). Every model under-predicts case counts at the true peak week specifically, by 42 to 76 percent at the median unit, more than it under-predicts the season total, which shows that the harder part of the forecasting problem is capturing acute intensity at the peak rather than accumulated burden over the season. The ensemble is a reasonable middle choice on every one of these summaries without being the best on any single one, consistent with it being chosen for robustness (Section 2.2) rather than for leading any one metric.

Table 7: Onset, peak, and total-burden accuracy, dengue state-level, four seasons and 26 states (104 units). Onset and peak-week errors are mean absolute error in weeks. Peak magnitude and total cases are the median log(predicted / observed) at the true peak week and over the full season respectively; 0 is unbiased, negative is under-prediction. Best value in each column is in bold.

Model	Onset MAE (weeks)	Peak-week MAE (weeks)	Peak magnitude (log-ratio)	Total cases (log-ratio)
Ensemble (conformal)	3.8	4.9	−0.71	−0.22
XGBoost	**3.7**	5.3	−0.55	−0.01
Climatological	4.2	**4.5**	−0.85	−0.32
GRU (deep learning)	4.1	5.0	−0.59	−0.12
Mechanistic	4.1	5.0	−1.43	−0.83
Seasonal-naive	4.2	4.8	−0.82	−0.30
Naive	4.9	17.4	−1.13	−0.14

Naive is the worst model by every measure in this table, including a 17-week peak-timing error; unlike in an earlier data vintage of this analysis, its total-cases figure no longer happens to look artificially competitive, which is itself a reminder that a single season-total number, like a single aggregate WIS, can be sensitive to incidental error cancellation and should not be read in isolation from the other columns.

3. Discussion

3.1. Robustness under regime shift

The most consequential finding is that model rankings are season-dependent and can invert in an outbreak year. A model chosen for its average score can be the least reliable in the season for which forecasts matter most, as the deep network is here. Two responses follow. One is to combine models, so that no single failure mode determines the forecast; the ensemble is never worst in any season and is best overall once recalibrated. The other is to prefer models whose uncertainty widens under distribution shift, as the mechanistic model's bootstrap-of-historical-seasons intervals do. These observations align with evidence from collaborative forecast hubs, where combined forecasts are consistently competitive with or better than their members [17–19]. The same holds for dengue specifically: across more than 180 locations worldwide, ensembles ranked consistently among the top performers at one-to-three-month horizons, including where individual models faltered [3]. Our result extends that to a full-season horizon and identifies the mechanism in this setting: the ensemble is never worst because its members fail in different seasons, and conformal recalibration then converts that robustness into the best overall score. The challenge exists to give Brazil's Ministry of Health a pre-season view of what the coming year may bring [1], and that use carries a concrete recommendation from these results: prefer a recalibrated ensemble to the single best-scoring model on a leaderboard. The season in which a leaderboard-topping model fails is, by construction, the season with no precedent in the training record, which is exactly the season a preparedness decision depends on most.

3.2. Metric choice is a reporting-standards issue

The identity of the best model changes with the aggregation of skill across a highly skewed set of states. This is not specific to our models or to Brazil. Any national forecasting evaluation with strong heterogeneity in population and incidence faces the same tension between magnitude-weighted and unit-weighted skill. Reporting a single aggregate obscures it. We recommend reporting both, as the interval-score and forecast-hub literature increasingly does [7], and we treat the normalized WIS the challenge adopts as the headline while retaining magnitude-weighted mean WIS and per-unit relative WIS alongside it.

3.3. Simple methods and the limits of complexity

Deep learning did not dominate, and simple methods, a climatological quantile model and a bootstrap of historical seasons, were hard to beat. This is consistent with forecasting-competition evidence that in small-sample, few-series regimes, parsimonious methods and combinations are strong [6, 24]. The pattern recurs one sprint earlier: in IMDC24, a Bayesian baseline model using no climate covariates at all was among the most consistently competitive submissions across states and seasons [1], echoing our own finding that a covariate-free climatological baseline and a historical-trajectory bootstrap are hard to beat with more heavily engineered alternatives. It recurs again against an architecturally close competitor: on monthly dengue counts in Freetown, Sierra Leone, a parsimonious autoregressive count model (INGARCH with a negative-binomial observation model) beat a bidirectional LSTM with a negative-binomial head, the closest published analogue of our own GRU, on distributional accuracy, with the deep model preferred only where tail reliability mattered most [5]. Three independent settings, two continents, and two surveillance systems give the same answer, which makes it a property of the forecasting regime rather than of any one pipeline. The same predecessor sprint offers an independent caution about tuning ensembles on limited past data: its CRPS-weighted logarithmic-pooling ensemble, with weights fit to 2023–2024 performance, badly overestimated the true 2025 outbreak in three of five states, because the models it weighted most heavily had performed well on those past seasons but poorly on the actual target season [1] – the same overfitting-to-past-folds failure mode our fold-1 tuning discipline (Section 2.8) exists to catch, now observed independently in another team's ensemble on a different edition of the same challenge. Our negative results sharpen the point. The covariate that helped the strongest 2024-sprint model, climate acting through a distributed-lag non-linear response (DLNM, Section 2.8 above) [15, 16], did not help our gradient-boosted forecaster, because that structure requires the regularized, partially pooled Bayesian formulation in which it was developed, whereas an unregularized tree over the same inputs overfits low-signal dimensions. The forecast-relevant climate information in our pipeline is the seasonal forecast and large-scale oscillation indices, not observed local weather. This clarifies where future modeling effort should go: toward a Bayesian spatiotemporal member that can use climate through an exposure-lag-response surface, rather than toward more covariates in a boosted model.

3.4. Conformal recalibration as transferable insurance

The conformal recalibration is simple, scale-free, and cheap, and it improves both accuracy and calibration. It connects to conformalized quantile regression [9] and to the broader literature on adaptive conformal methods for distribution shift [10, 31], which similarly recalibrate intervals in response to a changing data-generating process rather than assuming a fixed noise scale, though those methods update continuously as new data arrive whereas ours is calibrated once, on a single tuning season. Because it widens mainly the outer tails, it protects against the failure mode that dominates the outbreak season without materially degrading ordinary-season sharpness. For an operational forecaster who cannot know in advance whether the coming season will be ordinary or extreme, this asymmetric protection is the right default.

3.5. Limitations

The fourth backtest season is only partially resolved in the available data and is reported separately; its numbers will firm up as the 2025–26 season completes. Municipality-to-state aggregation discards sub-state epidemic asynchrony that a municipal model could exploit. The study covers two diseases in one country and is retrospective; we submitted the recalibrated ensemble to the challenge's forecast phase (the true 2026–27 season, EW41 2026 through EW40 2027) ahead of its 2026-09-10 deadline, but a genuine prospective score awaits that season's resolution. The comparison is single-team, so it does not capture the full diversity of approaches a multi-team sprint would, though our 26-state, four-track design covers more geographic and epidemiological ground than the inaugural 2024 sprint, which evaluated five highly endemic states and explicitly left lower-incidence, more recently affected regions untested [1]; the conformal factors are calibrated on one season under the challenge's tuning-fold discipline rather than by a fully online procedure. Finally, the 2023–24 outlier season we treat as a pure epidemiological surge may also reflect a data-quality confound: the 2024 season saw a substantial expansion of Oropouche fever, whose symptoms overlap with dengue's, and the inaugural sprint's organizers flag possible dengue-notification contamination from Oropouche misclassification as a contributing factor to that season's models failing so widely [1], a possibility our own pipeline cannot rule out with notification data alone.

3.6. Future work

The immediate next step is scoring the submitted forecast phase once the 2026–27 season resolves, which will provide a genuine prospective test of the recalibrated ensemble. Beyond it, four directions follow from our results and from the challenge organizers' own stated plans. First, a Bayesian spatiotemporal or endemic-epidemic member with a distributed-lag non-linear climate response [32, 33], which our negative result suggests is the right home for climate covariates. Second, hierarchical reconciliation across state, city, and national forecasts for coherence. Third, municipal-level feature engineering for the optional city tracks. Fourth, the organizers report moving their own collective ensemble toward a copula-based approach that samples coherent full-trajectory epidemic curves rather than combining marginal per-week quantiles independently; our per-quantile median ensemble does not guarantee this, but our mechanistic model's bootstrap of complete historical trajectories already provides trajectory-level coherence by construction, which is a reason to prefer it, or a copula-linked resampling of it, over a purely per-week combination when coherent full-season trajectories rather than marginal weekly intervals are the target.

4. Materials and methods

4.1. Pipeline overview

Fig 10 summarizes the full pipeline end to end before the following subsections detail each stage. Raw weekly surveillance and covariate data (Section 4.2, below) are filtered to each fold's own training cutoff, which removes any observation dated after that cutoff before anything downstream can see it (Section 4.3). For the machine-learning families, the cutoff-filtered history is expanded into a synthetic-origin training panel: every historical week becomes a candidate forecast origin, paired with the realized value at horizons 1 to 67 weeks ahead, which supplies enough training

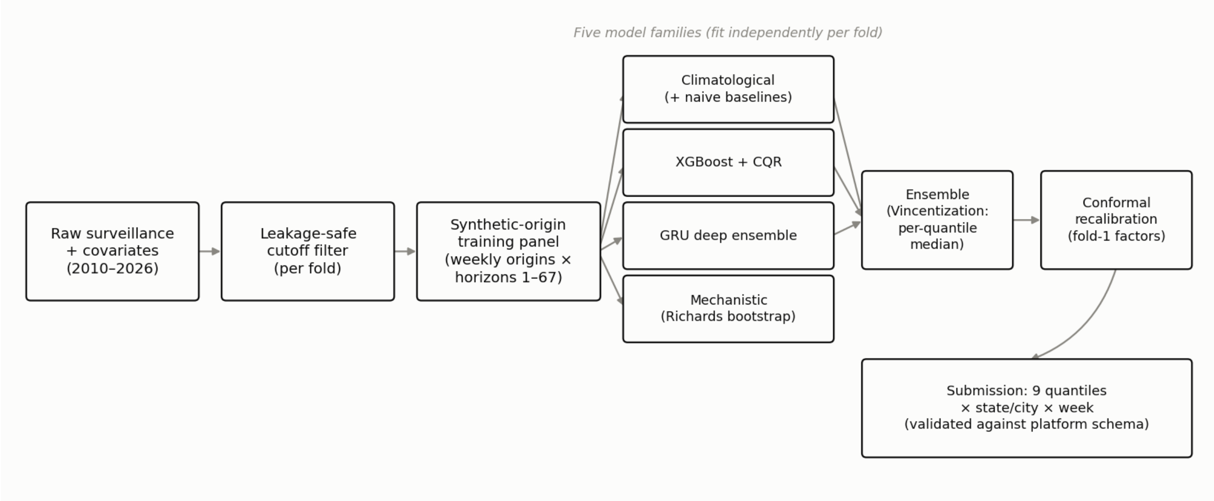

Figure 10: Pipeline overview. The dengue state-level pipeline, the most complex of the four tracks: cutoff-filtered data becomes a synthetic-origin training panel, four model families are fit independently, three of them (climatological, XGBoost, GRU) are combined by per-quantile median into an ensemble, and conformal recalibration produces the final submitted quantiles. The mechanistic model is evaluated for comparison but is not an ensemble member. Chikungunya-state and the city tracks (not shown) follow the same stages but submit a single family's output directly.

examples for gradient boosting and the deep network despite only four official evaluation windows existing. Four model families are then fit independently per fold: baselines and the climatological quantile model, a gradient-boosted model with conformalized quantile regression, the GRU deep ensemble, and the semi-mechanistic trajectory bootstrap. We evaluated two gradient-boosting implementations on the identical panel, LightGBM [11] and XGBoost [12], and adopted XGBoost, which scored significantly better (Section 2.4; Fig 5). The climatological, XGBoost, and GRU outputs (not the mechanistic model, which is evaluated separately for comparison) are combined by per-quantile median (Vincentization) into an ensemble, which is then conformally recalibrated using widening factors estimated once on a designated tuning season (Section 4.6). The output at every stage is the same canonical shape, nine quantiles per state or city and week, and the final recalibrated quantiles are validated against the platform's submission schema (non-negative, correctly nested, no date gaps) before being written out. Chikungunya-state and the two optional city tracks follow the same pipeline but submit a single family's output directly (LightGBM for chikungunya-state, unchanged from the dengue pipeline's original configuration, climatological for both city tracks; see Sections 2.9 and 2.10) rather than the three-member ensemble.

4.2. Data and targets

Weekly dengue and chikungunya case counts per municipality (2010–2026) are drawn from the Infodengue system, which harmonizes notifiable-disease surveillance [22], and aggregated to the 26 mandatory states, excluding Espírito Santo per the challenge. Covariates comprise weekly climate reanalysis (temperature, precipitation, humidity, pressure, thermal range, rainy days), a seasonal climate forecast, ocean-oscillation indices (El Niño–Southern Oscillation, Indian Ocean Dipole, Pacific Decadal Oscillation), yearly municipal population, and static climate-zone and biome covariates. The forecasting target is the weekly state case count over each season, submitted as the nine quantiles 0.025, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, and 0.975.

4.3. Backtest design and leakage control

The four official seasons are derived at runtime from the challenge's train and target flags. A confirmed 15-week gap separates each training cutoff from the start of its evaluation window, simulating reporting delay. Two leakage disciplines are enforced centrally in the harness rather than in individual models. All covariate access is filtered to dates at or before the training cutoff, and lag and rolling features are computed from the continuous cutoff-filtered history rather than from the gapped target window. Both are checked by automated regression tests, so a leak would fail the test suite rather than silently inflate scores.

4.4. Scoring

Forecasts are scored by the weighted interval score,

$$\mathrm{WIS}(F, y) = \frac{1}{K + \frac{1}{2}} \left[\tfrac{1}{2}\, |y - m| + \sum_{k=1}^{K} \frac{\alpha_k}{2}\, \mathrm{IS}_{\alpha_k}(\ell_k, u_k, y) \right],$$

over the four required central intervals, where m is the predictive median, IS_α is the interval score at level $1 - \alpha$ (the interval's width plus a penalty, scaled by $2/\alpha$, for however far the observation falls outside it), and the weights reproduce the mean pinball loss over the nine required quantiles, the exact discrete approximation to the continuous ranked probability score [7, 8]. Concretely, for a simplified two-interval version of the score with median $m = 10$, a 50 percent interval $[6, 14]$, a 90 percent interval $[2, 20]$, and an observation $y = 15$: the median term is $\frac{1}{2}|15 - 10| = 2.5$; the 50 percent interval ($\alpha = 0.5$) is exceeded by 1 (since $15 > 14$), giving $\mathrm{IS}_{0.5} = (14-6) + \frac{2}{0.5}(15-14) = 12$, weighted by $\alpha/2 = 0.25$ to 3; the 90 percent interval ($\alpha = 0.1$) is not exceeded, giving $\mathrm{IS}_{0.1} = (20 - 2) = 18$, weighted by 0.05 to 0.9; so, with $K = 2$ intervals, $\mathrm{WIS} = (2.5 + 3 + 0.9)/(2 + 0.5) = 2.56$. Our implementation is verified against this hand-computed value, against an independent pinball-loss re-derivation, and against the challenge platform's own scorer. We also report empirical interval coverage, a decomposition of WIS into dispersion and directional (over- and under-prediction) penalties, a scale-free per-unit relative WIS, and the normalized WIS (sum of WIS divided by sum of observed cases) that the challenge adopts as its headline metric.

4.5. Models

Baselines. A naive model (last observed value with empirical horizon-difference quantiles), a seasonal-naive model (same-epidemiological-week history), and a climatological quantile model (direct order-statistic quantiles per state and epidemiological week over a five-week window).

Gradient boosting. Only four official evaluation windows exist per disease, far too few to fit a tree ensemble, so a gradient-boosted quantile model is instead fit on a synthetic-origin panel: within each fold's own cutoff-filtered history, every Sunday from roughly 2014 onward becomes a candidate forecast origin, and every (origin, horizon) pair from 1 to 67 weeks ahead becomes one training row, with origin-anchored lag, rolling, seasonal, climate, ocean, seasonal-forecast, and static features as inputs and the realized case count that many weeks later as the label. For example, one row might read: origin = the first Sunday of March 2021, horizon = 12 weeks, features computed using only data available on that March Sunday, label = the observed case count 12 weeks after it. This turns roughly 500 usable weekly origins per state, crossed with up to

67 horizons, into tens of thousands of training rows for a single fold, while evaluation still only ever happens at that fold's own four-times-a-year official target window. We evaluated two gradient-boosting implementations on this identical panel: LightGBM [11], one booster per quantile level, and XGBoost [12], a single booster with a native multi-quantile objective covering all nine levels jointly. XGBoost scored significantly better both overall and on the outlier season (Sections 2.2 and 2.4) and is the version used in the main ensemble; LightGBM remains the gradient-boosted family used for the chikungunya-state track, on which it was not re-evaluated against XGBoost. Predictive intervals for both are calibrated by conformalized quantile regression (CQR) [9]: a held-out slice of the training panel is used to measure how much each quantile needs to be shifted so that it achieves its nominal coverage, and that shift is then applied to the model's raw quantile predictions, in the same spirit as, but computed independently from, the ensemble-level conformal recalibration of Section 4.6.

Deep learning. A global gated recurrent unit (GRU) network [13], a type of recurrent neural network designed to retain information over many time steps, with per-state embeddings and a negative-binomial output head (chosen over a Poisson head because case counts are more variable than a Poisson distribution allows), trained as a deep ensemble of independently seeded runs [14]; quantiles are analytic, meaning computed directly from the fitted negative-binomial distribution's formula rather than by simulation, and pooled across ensemble members by quantile averaging.

Semi-mechanistic. A model that bootstraps historical season trajectories under a negative-binomial observation model. Trajectories come from a local reimplementation of Richards-curve epidemic-scanner fits [26]: the Richards curve is a flexible, S-shaped growth-then-plateau curve long used to summarize an epidemic's rise and fall, fit separately to each historical season, from which we resample plausible future trajectories. Because it draws on the full spread of past seasons rather than assuming a fixed noise scale, this produces wide, well-calibrated intervals under distribution shift.

Ensembles. Models are combined by per-quantile median (Vincentization) and, separately, by inverse-WIS weights: each model's weight is proportional to $1/\overline{\text{WIS}}$ on the designated tuning season, so a model that scored better on that season counts for more, tuned only on the designated tuning season, so weights are never fit on the seasons on which the ensemble is reported.

4.6. Conformal recalibration

Section 2.3 gives a worked numeric example of this procedure's effect on one interval; here we give the full definition. For each central interval level L with bounds $[\ell_L, u_L]$ and median m, we compute a multiplicative widening factor s_L that makes the interval reach its nominal coverage on the tuning season. For each calibration unit, the multiple needed to just cover the observation is $r = (y - m)/(u_L - m)$ when $y > m$ and $r = (m - y)/(m - \ell_L)$ when $y < m$; s_L is the $L/100$ empirical quantile of r, a scale-free, multiplicative analogue of conformalized quantile regression that is robust to the large cross-state differences in scale. The interval is then widened about the median by s_L, and monotonicity across quantiles is restored by a rearrangement step. Factors are estimated on the tuning season and applied to all seasons; recalibrating the ensemble output outperforms recalibrating any single member. The same frozen, fold-1-derived factors are reused unchanged for the forecast phase, consistent with fitting recalibration weights only once and never on the season being forecast.

4.7. Reproducibility

All code, the backtesting harness, per-state scored predictions, and validated submission files for four official tracks (dengue and chikungunya, state and city level) are released. The pipeline is deterministic under a fixed seed, and an automated test suite covers leakage safety, scoring correctness, model persistence, and every model family. A provenance manifest records the exact data and code state behind each reported number. The Supporting Information provides the full feature list and frozen hyperparameters, the deep-learning architecture, the mechanistic derivation, per-state and per-fold tables, the ensemble-member and weight robustness analysis, and an item-by-item EPIFORGE 2020 reporting checklist [20].

Data and code availability

The pipeline was developed by team Neural Earth for the 3rd Infodengue–Mosqlimate Dengue Challenge. All code, data-processing pipeline, backtests, figures, and validated submissions are openly available at `https://github.com/kamrul28890/3rd_imdc_purdue_neuralearth`. The surveillance and covariate data are distributed by the challenge through the Mosqlimate platform.

Author contributions

Conceptualization, methodology, software, formal analysis, investigation, visualization, and writing of the original draft: M.K.K. Validation, data curation, and writing (review and editing): E.J.E. and A.A.H. All authors read and approved the final manuscript.

Competing interests

The authors have declared that no competing interests exist.

Funding

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Ethics statement

This study used aggregate, publicly available disease-surveillance and covariate data distributed through the Infodengue and Mosqlimate platforms. No individual-level data were used and no ethical approval was required.

AI usage

Development of the analysis pipeline, statistical methods, code, and portions of this manuscript was assisted by Claude (Anthropic), an AI assistant, under the direction and review of the human authors. All methodological decisions, interpretations, and conclusions are the responsibility of the human authors, who reviewed and verified the code, analyses, and results.

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